

D794

Computer Architecture and Systems

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A microcontroller keeps separate memories and buses for instructions and data, so it can fetch the next instruction while reading data. The design team is based in Austin. Which architecture is this?

- A. RISC architecture.
- B. It depends on where the team is based.
- C. Von Neumann architecture.
- D. Harvard architecture.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A program executes 2 billion instructions with an average CPI of 1.5 on a 3 GHz processor. The machine has 16 GB of RAM. What is the CPU time?

- A. 1.0 second
- B. About 0.44 seconds
- C. It depends on the 16 GB of RAM.
- D. About 0.67 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A program runs 4 billion instructions at a CPI of 2.0 and must finish in 2 seconds. What is the minimum clock rate?

- A. 16 GHz
- B. 2 GHz
- C. 1 GHz
- D. 4 GHz

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A cache has a 1 ns hit time, a 5% miss rate, and a 100 ns miss penalty. The server has 64 cores. What is the average memory access time?

- A. 96 ns
- B. 6 ns
- C. 384 ns
- D. 5 ns

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A cache has a 2 ns hit time and an 80 ns miss penalty. What miss rate gives an average memory access time of 6 ns?

- A. 2.5%
- B. 95%
- C. 7.5%
- D. 5%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A phone-chip designer wants simple, fixed-length instructions that mostly finish in one cycle, with memory reached only through load and store instructions, to save power. The phone will have a large battery. Which approach fits?

- A. Harvard architecture.
- B. RISC.
- C. CISC, since a large battery makes power irrelevant.
- D. CISC.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In the mid-2000s, processor clock speeds stopped rising quickly even though transistor counts kept growing, and chipmakers began putting several cores on each chip. A popular game console launched around then. Why did this shift happen?

- A. Transistor counts stopped increasing.
- B. Game console makers demanded more cores.
- C. Higher clocks produced too much power and heat, so extra transistors went into more cores.
- D. Software no longer needed more speed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A server's CPU spends 40% of its time checking a network card in a loop for packets that arrive at irregular times. The card is from a well-known vendor. Which change frees the CPU?

- A. Poll the card more often.
- B. Swap in another vendor's card.
- C. Use interrupts so the card signals on arrival, with DMA moving data into memory.
- D. Raise the CPU's clock speed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A sensor sends the 32-bit value 0x12345678 in big-endian order to an x86 server, which is little-endian and reads the raw bytes straight into an integer. The link runs at 1 Gbps. What value does the server read?

- A. 0x78563412
- B. 0x87654321
- C. It depends on the link speed.
- D. 0x12345678

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Two devices with different byte orders exchange binary integers. What is the standard fix?

- A.** Agree on a defined byte order for the wire, such as network byte order, and convert on send and receive.
- B.** Nothing; byte order never matters for integers.
- C.** Reverse the bits within each byte.
- D.** Send the values as floating-point numbers.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A 5-stage pipeline has 2 ns per stage. The chip was made on a 5 nm process. With no stalls, how long does it take to complete 100 instructions?

- A.** 1,000 ns
- B.** 10 ns
- C.** 208 ns
- D.** 200 ns

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A processor is redesigned from a single-cycle design into an ideal 5-stage pipeline. What happens to one instruction's latency and to overall throughput?

- A.** Throughput is always exactly 5 times higher in real programs.
- B.** Latency stays about the same; throughput rises up to about 5 times.
- C.** Both latency and throughput improve 5 times.
- D.** Each instruction finishes about 5 times faster.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A microcontroller keeps separate memories and buses for instructions and data, so it can fetch the next instruction while reading data. ~~The design team is based in Austin.~~ Which architecture is this?

Target word: name the architecture **Decides:** Separate instruction and data paths.

A	Adjacent category	RISC describes the instruction set, not memory paths.
B	Camouflage	Location has nothing to do with architecture.
C	Confusion set	Von Neumann shares one memory and bus.
D	Correct	Separate memories allow both fetches at once.

2. Answer A Concept 2

A program executes 2 billion instructions with an average CPI of 1.5 on a 3 GHz processor. ~~The machine has 16 GB of RAM.~~ What is the CPU time?

Target word: CPU time **Decides:** Instructions $\times$ CPI $\div$ clock rate.

A	Correct	$2 \times 10^9 \times 1.5 / (3 \times 10^9) = 1.0$ s.
B	Inverted operation	Divides by CPI instead of multiplying.
C	Camouflage	The equation needs no memory size.
D	Dropped element	Ignores CPI: $2 \times 10^9 / 3 \times 10^9$.

3. Answer D Concept 2 · transfer

A program runs 4 billion instructions at a CPI of 2.0 and must finish in 2 seconds. What is the minimum clock rate?

Target word: solve for clock rate **Decides:** Clock = instructions $\times$ CPI $\div$ time.

A	Confusion set	Multiplies by the time instead of dividing.
B	Dropped element	Ignores CPI.
C	Inverted operation	Divides by CPI instead of multiplying.
D	Correct	$4 \times 10^9 \times 2 / 2 = 4 \times 10^9$ cycles per second.

4. Answer B Concept 3

A cache has a 1 ns hit time, a 5% miss rate, and a 100 ns miss penalty. ~~The server has 64 cores.~~ What is the average memory access time?

Target word: AMAT **Decides:** $1 + 0.05 \times 100$.

A	Inverted operation	Uses the 95% hit rate as the miss rate.
B	Correct	$1 + 5 = 6$ ns.
C	Camouflage	Multiplies by the core count.
D	Dropped element	Leaves out the hit time.

5. Answer D Concept 3 · transfer

A cache has a 2 ns hit time and an 80 ns miss penalty. What miss rate gives an average memory access time of 6 ns?

Target word: solve for miss rate **Decides:** $(6 - 2) \div 80$.

A	Wrong stem number	2 / 80 uses the hit time as the numerator.
B	Inverted operation	That is the hit rate.
C	Dropped element	6 / 80 drops the hit time.
D	Correct	4 / 80 = 0.05.

6. Answer B Concept 4

A phone-chip designer wants simple, fixed-length instructions that mostly finish in one cycle, with memory reached only through load and store instructions, to save power. ~~The phone will have a large battery.~~ Which approach fits?

Target word: instruction-set style **Decides:** Simple, fixed-length, load/store.

A	Adjacent category	A memory organization, not an instruction-set style.
B	Correct	Those are RISC traits, favored for power efficiency.
C	Camouflage	Battery size does not change the design goal.
D	Confusion set	CISC uses complex, variable-length instructions.

7. Answer C Concept 5

In the mid-2000s, processor clock speeds stopped rising quickly even though transistor counts kept growing, and chipmakers began putting several cores on each chip. ~~A popular game console launched around then.~~ Why did this shift happen?

Target word: cause of the multicore shift **Decides:** Transistors grew; clocks stalled.

A	Confusion set	The stem says transistor counts kept growing.
B	Camouflage	A product launch did not cause the physics limit.
C	Correct	The power wall limited clock gains.
D	Adjacent category	Demand for performance continued.

8. Answer C Concept 6

A server's CPU spends 40% of its time checking a network card in a loop for packets that arrive at irregular times. ~~The card is from a well known vendor.~~ Which change frees the CPU?

Target word: reduce I/O overhead **Decides:** Polling for irregular events.

A	Inverted operation	Spends even more CPU time checking.
B	Camouflage	The vendor is not the cause; the polling design is.
C	Correct	The CPU works only when data arrives and does not copy each byte.
D	Adjacent category	Still wastes the same share of time polling.

9. Answer A Concept 7

A sensor sends the 32-bit value 0x12345678 in big-endian order to an x86 server, which is little-endian and reads the raw bytes straight into an integer. ~~The link runs at 1 Gbps.~~ What value does the server read?

Target word: value read with mismatched byte order **Decides:** Bytes are reversed, not bits or nibbles.

A	Correct	Bytes 12 34 56 78 read little-endian give 0x78563412.
B	Confusion set	Reverses digits (nibbles), not bytes.
C	Camouflage	Speed does not change byte order.
D	Dropped element	Ignores the byte-order mismatch.

10. Answer A Concept 7 · transfer

Two devices with different byte orders exchange binary integers. What is the standard fix?

Target word: cross-platform data exchange **Decides:** Agree on one wire format.

A	Correct	Each side converts to and from one agreed format.
B	Absolute language	Multi-byte integers are exactly what byte order affects.
C	Confusion set	Endianness is about byte order, not bit order.
D	Adjacent category	Floats have the same byte-order issue.

11. Answer C Concept 8

A 5-stage pipeline has 2 ns per stage. ~~The chip was made on a 5-nm process.~~ With no stalls, how long does it take to complete 100 instructions?

Target word: pipelined execution time **Decides:** $(k + n - 1)$ cycles.

A	Confusion set	That is the time without pipelining.
B	Wrong stem number	One instruction's latency.
C	Correct	$(5 + 99) \times 2 \text{ ns} = 208 \text{ ns}$.
D	Dropped element	Drops the cycles needed to fill the pipeline.

12. Answer B Concept 8 · transfer

A processor is redesigned from a single-cycle design into an ideal 5-stage pipeline. What happens to one instruction's latency and to overall throughput?

Target word: latency vs. throughput **Decides:** Pipelining overlaps instructions; it does not speed each one.

A	Absolute language	Hazards and stalls keep real gains below the ideal.
B	Correct	More instructions are in flight, each taking as long as before.
C	Confusion set	Only throughput improves.
D	Inverted operation	Reverses which measure improves.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Von Neumann vs. Harvard	Von Neumann shares one memory and bus for instructions and data; Harvard separates them so both can be fetched at once.	Mixing up which design shares the bus.	Separate or shared paths for instructions and data.
2	CPU performance equation	$\text{CPU time} = \text{instruction count} \times \text{cycles per instruction} \div \text{clock rate}$.	Ignoring CPI or treating clock speed alone as performance.	Instruction count, CPI, and clock rate in one stem.
3	Memory hierarchy and AMAT	Average memory access time = hit time + miss rate $\times$ miss penalty.	Leaving out hit time or using the hit rate as the miss rate.	A hit time, a miss rate, and a miss penalty.
4	RISC vs. CISC	RISC uses simple, fixed-length instructions and load/store memory access; CISC uses complex, variable-length instructions that can work on memory directly.	Thinking more complex instructions always mean faster, more efficient chips.	Instruction length, complexity, and power goals.
5	The power wall and multicore	Rising power and heat stopped fast clock-speed gains, so extra transistors went into multiple cores.	Believing transistor growth stopped.	Clock speeds flatten while core counts rise.
6	Interrupts, polling, and DMA	Interrupts let devices signal the CPU only when needed, and DMA moves data without the CPU copying each byte; polling wastes cycles for irregular events.	Polling faster to reduce overhead.	A CPU busy checking a device in a loop.
7	Byte order (endianness)	Big-endian stores the most significant byte first, little-endian the least; systems exchanging binary data must agree on an order and convert.	Reversing bits or nibbles instead of bytes, or ignoring order.	Raw multi-byte values moving between different platforms.
8	Pipelining	Pipelining raises throughput (about one instruction per cycle once full) but does not shorten one instruction's latency; n instructions on k stages take $(k + n - 1)$ cycles.	Expecting each instruction to finish k times faster.	A stage count, stage time, and instruction count.

Course scope (public description): Computer Architecture and Systems offers a comprehensive examination of the design and development of computer systems. Students will explore the evolution of computer architecture, understanding how computers have advanced and changed over time. The course covers different layouts and designs used in both hardware and software solutions, helping students to grasp the complex structures of computer systems. Students will learn how to develop detailed plans for implementing computing solutions that work across various platforms. The course likewise covers how to integrate various devices to ensure they operate and communicate smoothly with each other.